



RossClean BMC

Terminal disinfectant cleaner

RossClean BMC is a high-performance blend of surface-active biocide, surfactant, alkaline builders and sequestering agents. It's a broad spectrum disinfectant suitable for open plant cleaning applications throughout the food, beverage and dairy industries.

Product is ideal for high care application areas. Its unique formulation provides excellent disinfectant activity against most vegetative forms of micro-organisms including gram-positive and gram-negative bacteria and yeast and molds.

The ingredients used in RossClean BMC are carefully selected for reduced environmental impact. Product is suitable for manual, soak or spray application and for fogging use.

RossClean BMC is recommended for application in all open plant cleaning processes. It should be applied after surfaces have been thoroughly cleaned and rinsed. It can be used for disinfecting floors, walls, utensils and other food preparation equipment and is suitable for use in Dairy, meat and poultry processors, snack foods, dairies, beverage plants and most other types of food processing operations.

Key Benefits

- Highly effective disinfectant for food & dairy industry use
- Single step cleaning
- Non-corrosive
- Free-rinsing
- Non-tainting, use on food contact surfaces
- Effective with both soft and hard water

Application Method

Use product at concentration of between 0.2-3% v/v depending on application. Always rinse thoroughly after use.

For more details, please contact your local Buzil Rossari representative.

Technical Data

Appearance	Colorless liquid
pH (1% solution at 20°C)	10.5 - 11.5
Solubility	High solubility in water
Density at 25°C	1 g/ml (+/- 5%)



RossClean BMC

Terminal disinfectant cleaner

Safety Assessment

Eye contact: Flush with water for at least 15 minutes. If irritation occurs and persists, obtain medical attention.

Skin contact: Wash with plenty of soap and water. If irritation occurs and persists, obtain medical attention. Wash clothing before reuse.

Inhalation: Get fresh air. If difficulty in breathing or discomfort occurs, call a physician.

Ingestion: Drink one to two glasses of water. Do not induce vomiting. Do not give anything by mouth to an unconscious individual. Call a physician immediately.

Storage and Handling

All safety information on the handling and use of this product is provided in the Safety Data Sheet which can be consulted and / or obtained from our representatives. Keep this product in the original packaging or, where possible, in a tank of bulk reception, protected from extreme temperatures.

Declaration

Technical figures given in this data sheet are typical values based on internal laboratory measurements, following appropriate International Standard Methods, and should not be taken as a warranty or representation. It is the end user's responsibility to make sure that the product matches its requirements.

For more information on this product please contact your local Buzil Rossari representative.

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For more information on this product please contact your local Buzil Rossari representative.

Buzil Rossari Private Limited

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MATERIAL SAFETY DATA SHEET

according to REGULATION (EU) No 453/2010, annex I

RossClean BMC

Date of issue: 08.06.2018

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1. Identification of the mixture and of the company / undertaking

1.1. Product identifier

RossClean BMC

1.2. Relevant identified uses of the mixture and uses advised against

Professional terminal disinfectant cum cleaner for open plant applications for food & beverage industries.

1.3. Details of the supplier of the safety data sheet

Manufacturer:

Manufactured under licence from buzil
ROSSARI BIOTECH LIMITED
Plot No. 10 & 11, Survey No.90/1/10 & 90/1/11/1,
Kumbharwadi, Village Naroli
Silvassa-396235
Dadra and Nagar Haveli (UT)
India

e-mail info@rossarimail.com

www.buzil-rossari.com & www.rossari.com

1.4. Emergency telephone number

+91-22-61233800/61233887

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2. Hazards identification

Risk phrases :

Hazards for man and the environment :

R 22 Harmful if swallowed

R 34 - Causes burns

R 36 - Irritating to eyes

R 50 - Very toxic to aquatic organisms



Classification system : The classification complies with current EC regulations on dangerous substances and preparations..

3. Composition / information on ingredients

Chemical characterization

Description : Mixture in water of non-dangerous ingredients and the substances listed below.

Hazardous ingredients :

Alkyldimethyl benzylammonium chloride C,N; R 22-34-50 10-15%

Didecyl Dimethyl Ammonium Chloride Xi; <5%

Biguanide Xi; <2%

Ingredients according to EC Detergents Regulation 648/2004

non-ionic surfactants <5%

Na salt of ethylene diamine tetra acetic acid <5%

Additional information : The full text of the relevant risk phrases is given in section 16..

4. First aid measures

General information : No special measures required.

Inhalation : Remove from source of exposure. Obtain medical attention if symptoms develop.

Skin contact : Wash skin immediately and thoroughly with water. Remove contaminated clothing.

Obtain medical attention if symptoms develop.

Eye contact : Wash immediately with copious amounts of water. Obtain medical attention if symptoms develop.

Ingestion : Rinse out mouth and then drink one or two glasses of water(or milk). Obtain medical attention.



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5. Firefighting measures

Suitable extinguishing media : CO₂, extinguishing powder or water jet. Fight larger fires with water jet or alcohol resistant foam.

Protective equipment : No special measures required

6. Accidental release measures

Protection of personnel : Not required

Environmental precautions : Dilute with plenty of water

Method of disposal : Absorb with liquid-binding material (sand, diatomite, universal binders, sawdust).

Dispose off the collected material according to the regulations.

7. Handling and storage

Handling (see also sections 8 and 15)

Information for safe handling : Use common rules for working with chemicals.

Information about protection against explosions and fires : No special measures required.

Storage requirements for storage rooms and containers : According to local legislation.

Storage in one common storage facility : According to local legislation.

Further information about storage conditions : None.

8. Exposure controls / personal protection

Additional information about design of technical systems : No further data; see section 7.

Components with limit values that require monitoring at the workplace :

Additional information : The lists that were valid during the compilation were used as basis.

General protective and hygienic measures :

Keep away from foodstuffs, beverages and food. Take off immediately all contaminated clothing.

Wash hands during breaks and at the end of work. Do not inhale gases /fumes /aerosols. Avoid contact with the eyes and skin.

Breathing equipment : Personal protection is not normally required. However, inhalation of dust, vapour or aerosols should be avoided.

Protection of hands : Use safety gloves.



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Eye protection: Safety glasses recommended during handling the undiluted product that involves a risk of splashing.

9. Physical and chemical properties

Information on basic physical and chemical properties

Appearance	Clear liquid
Odour	Characteristic
pH-value(1%)	10.5 – 11.5
Melting point	ND
Boiling point	ND
Density (25 °C)	1.0 - 1.1 gm/cm ³
Solubility in water	Miscible in water
Viscosity (25 °C)	ND
Specific gravity	ND

10. Stability and reactivity

Thermal decomposition / conditions to be avoided : No decomposition if used according to specifications.

Dangerous reactions : None.

Hazardous decomposition products : None.

11. Toxicological information

Information on toxicological effects

Acute toxicity:

Primary irritant effect:

Skin contact : Causes irritation

Eye contact : Causes severe irritation

Inhalation : Causes irritation

Ingestion : Causes irritation

12. Ecological information

General notes : Must not reach sewage water or drainage ditch undiluted or unneutralized. When used for its intended purpose this product should not cause adverse effects in the environment.



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The information in this section refers to the undiluted product.

13. Disposal considerations

Product recommendation : Must not be disposed of with domestic waste. Do not allow undiluted product to reach sewage system.

Uncleaned packaging recommendation : Dispose of observing official regulations.

Recommended cleaning agent : Water, if necessary with cleaning agent.

14. Transport information

Land transport ADR/RID (cross-border)

ADR/RID Class : 8 Corrosive substances

Kemler Number : 80

UN-Number : 3267

Packaging Group : III

Label : 8

Proper Shipping Name : 3267 CORROSIVE LIQUID, BASIC, ORGANIC, N.O.S.

ALKYLDIMETHYLBENZYLAMMONIUMCHLORIDE

Maritime transport IMDG :

IMDG Class : 8

UN-Number : 3267

Label : 8

Packaging Group : III

EMS Number : 8-15

Maritime Pollutant : Yes

Proper Shipping Name : 3267 CORROSIVE LIQUID, BASIC, ORGANIC, N.O.S.

ALKYLDIMETHYLBENZYLAMMONIUMCHLORIDE

Air transport ICAO-TI and IATA-DGR :

ICAO / IATA Class: 8

UN-Number : 3267

Label : 8

Packaging Group : III



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Proper Shipping Name : 3267 CORROSIVE LIQUID, BASIC, ORGANIC, N.O.S.
ALKYLDIMETHYLBENZYLAMMONIUMCHLORIDE

15. Regulatory information

Classification according to EC guidelines : The product has been classified and labeled in accordance with current EC Directives.

Symbol and hazard designation of product:



16. Other information

The information in this document is based on our best present knowledge. However, it does not constitute a guarantee or any specific product features and does not establish a legally binding contract.



Technical Specification

BMC Cleaning Chemical: RossClean BMC (Upgrade)

Active Ingredients Details

Sr. No	Particular	% Composition
1	Poly Hexa Methylene Biogunide (0.4%)	< 2 %
2	Di Decyl Di Methyl Ammonium Chloride (0.4%)	< 5%
3	Benzyl Konium Chloride	10 – 15 %
4	Activators	< 5%

Physical/Chemical Properties

Sr. No	Particular	Values
1	Appearance: Colorless white to pale yellow	Colorless
2	Odor: Odorless	Odorless
3	State: Liquid	Liquid
4	Solubility in water: complete	Complete
5	PH: >10.0	8 ± 1
6	Boiling Point: NA	NA
7	Flash Point: Non-combustible	NA
8	Density @20°C : 1.05 to 1.16	1.00 – 1.10
9	Stability: Stable under ordinary condition during use / storage	Stable
10	Hazardous Polymerization: will not occur	Does not



Chemical Compatibility Chart

1. DISCUSSION

These tables list the chemical compatibility chart of various metallic materials as used in RossClean BMC products at ambient temperatures, 77°F (25°C). This data has been compiled from literature published by various material suppliers. It is given as background information only. In any given case, factors such as concentration, temperature, degree of agitation and presence of impurities influence the rate of corrosion. The user, through his own testing and evaluation, must determine the suitability of the material for the application.

Ingredients	Aluminium	Plastic	Stainless Steel
Di Decyl Di Methyl Ammonium Chloride (0.4%)	A	A	A
Benzyl Konium Chloride (12-15%)	A	A	A
Poly Hexa Methylene Biogunide (0.4%)	A	A	A
Activators(For elimination of Bacteria's) (0-5%)	A	A	A

2. GUIDE TO COMPATIBILITY KEY

A : Acceptable

Q : Questionable – The Bal Seal material has limited resistance to this chemical; use of the material depends on the conditions of the application.

NR : Not Recommended – The Bal Seal material is not compatible with this chemical; its use is not recommended.

NA : Not Available – Chemical compatibility information is not available at this time.

3. CHEMICAL COMPATIBILITY

RossClean BMC when applied at the recommended concentration and temperature is suitable for use on the grades of plastic, stainless steel, aluminium, etc. commonly found in the processed food industry.



Test Method to Check Presence of Chemical Residue in Milk

Objective

To facilitate the milk collection personnel at the bulk milk cooler (BMC) centres to test the QAC residue in milk.

Method

- To a centrifuge tube add 1mL of milk, 5mL of water, 1mL of EOSIN solution and 2mL buffer and shake for 10 secs
- Centrifuge for 5 minutes at 3200 RPM
- If QAC is present the bottom layer assumes a RED or PINK
- Samples containing 1mg/kg of QAC shows a faint PINK
- If the colour is deep RED or PINK, the amount of QAC can be approx.. determined by titration with a standard anionic detergent solution



Standard Analytical Activity Test Method

Product	RossClean BMC	Issued On	01/05/2018
Application	Terminal disinfectant for open plant applications	Next Review On	01/05/2023

AIM

Determination of active content in RossClean BMC

APPARATUS

- Volumetric flask 500 ml
- Stoppard graduated cylinder – 100 ml
- Burette with stand - 25 ml
- Pipette–10 ml
- Glass Beaker - 250 ml

REAGENTS

- 0.004 M Sodium dodecyl sulfate solution
- Methylene blue solution
- Chloroform (AR grade)
- Distilled Water

PROCEDURE

- Weight 0.5 gm of sample in 250 ml glass beaker
- Add 100 ml Distilled water & stir vigorously to dissolve the sample
- Pipette out 10ml of this solution in 100ml Stoppard graduated cylinder
- Add it 15 ml chloroform & 25 ml methylene blue solution & shake vigorously
- Titrate this solution against 0.004 M Sodium dodecyl sulfate solution. End point will be that at which the colour intensity in both the phase is same when viewed under standard condition of light.

CALCULATION

Active matter = $(B.R * 100) / 5$

B.R : Burette Reading



Microbial / Minimum Inhibitory Concentration Test

Product	RossClean BMC
Application	Terminal Disinfectant cum Cleaner for Open Plant Applications
Date of Analysis	25-10-2018
Date of Report	29-10-2018

Sr. No.	Active Concentration	UOM	Tube Observation
1	45000	µg/ml	No turbidity
2	22500	µg/ml	No turbidity
3	11250	µg/ml	No turbidity
4	5625	µg/ml	No turbidity
5	2813	µg/ml	No turbidity
6	1406	µg/ml	No turbidity
7	703	µg/ml	No turbidity
8	352	µg/ml	No turbidity
9	Positive Control	µg/ml	Turbidity
10	Negative Control	µg/ml	No turbidity
11	Broth Control	µg/ml	No turbidity
12	SAMPLE	µg/ml	----

Calculation

$$\text{MIC} = \frac{(\text{Lowest Conc of sample inhibiting growth} + \text{Highest Conc. Sample allowing the growth})}{2}$$

MIC = Less than 351 µg/ml OR 0.000351 g/ml

Remarks

The MIC of the given sample is below 0.000351 g/ml

Declaration

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Empty HDPE 2 Container Disposal Method

Do not re-use empty containers. Non-refillable container. Clean container promptly after emptying. Rinse as follows: Empty the remaining contents into application equipment or a mix tank. Fill the container ¼ full with water and recap. Shake for 10 seconds. Pour rinsate into application equipment or a mix tank or store rinsate for later use or disposal. Drain for 10 seconds after the flow begins to drip. Offer container for recycling, to recyclers having valid registration.

Dispose in compliance with all Central, State, Gram Panchayat and local laws and regulations.

As per the notification Published in the Gazette of India, Part-II, Section-3, Sub-section (i) of Ministry of Environment, Forest and Climate Change, New Delhi, the 18th March, 2016, as amended 27th March, 2018.

Responsibility of Gram Panchayat. –

Every gram panchayat either on its own or by engaging an agency shall set up, operationalise and co-ordinate for waste management in the rural area under their control and for performing the associated functions, namely, -

- a. ensuring segregation, collection, storage, transportation, plastic waste and channelization of recyclable plastic waste fraction to recyclers having valid registration; ensuring that no damage is caused to the environment during this process;
- b. creating awareness among all stakeholders about their responsibilities; and
- c. ensuring that open burning of plastic waste does not take place





Storage condition and Shelf life of the chemical

Storage

Requirements to be met by storerooms and receptacles:

- Avoid storage near extreme heat, ignition sources or open flame
- Unsuitable material for receptacle: aluminium
- Unsuitable material for receptacle: steel

Information about storage in one common storage facility:

- Store away from foodstuffs
- Store away from oxidizing agents
- Store away from reducing agents
- Do not store together with acids

Further information about storage conditions:

- Keep container tightly closed
- Protect from freezing

Shelf Life

24 Months from the date of manufacturing.

Packaging

2 x 5 Litre Cans



BMCU Cleaning Standard Operating Procedure

Automated CIP Cleaning Procedure for BMC Units

Objective

To facilitate the milk collection personnel at the bulk milk cooler (BMC) centres to follow the correct procedure for collection of safe and superior quality milk in the shortest possible time.

Automated Cleaning-in-place (CIP)

1. Thoroughly rinse the interior of the BMC tank immediately after unloading milk into tankers, with normal water. Drain the rinse water.
2. Prepare a **0.3-0.5% RossClean BMC** cleaning solution in hot water (30-40°C) in the balance tank.
3. Circulate this solution through the pipelines and the milk hose for about 10-15 minutes to clean the pipelines and drain.
4. Spray hot water (70-75°C) on the interior surface of the BMC tank; circulate through the pipelines/milk hose for 10-15 minutes and drain.

OR

Spray/Circulate cold sanitizer **Prudent 15** solution (0.1-0.15%) BMC tank for 5-7 minutes and drain.

OR

Through rinse with portable RO water for 10-15 minutes and drain.

5. Ensure No water is sprayed after cleaning and sanitization.



Manual Cleaning Procedure for BMC Units

Objective

To facilitate the milk collection personnel at the bulk milk cooler (BMC) centres to follow the correct procedure for collection of safe and superior quality milk in the shortest possible time.

Manual Cleaning

1. Clean the system immediately after pumping the milk into the tanker. Also clean the system whenever contamination occurs or is suspected.
2. Ensure that the milk tank is empty and the refrigeration unit and the milk agitator are in the 'OFF' position during cleaning.
3. Rinse the interior of the BMC tank and the balance tank immediately after pumping all the milk into a tanker, by spraying lukewarm water of about 50°C through a hose pipe and re-circulate this water thoroughly to rinse the pipelines and the milk hose using a pump. Drain the rinse water.
4. Scrub the interior of the BMC tank and the balance tank thoroughly with **0.3% RossClean BMC** solution using a brush. Clean the outlet connection and outlet valve manually. Drain this water.
5. Dismantle all the parts like milk pump including the impeller, impeller casing, piping, valves, inline strainer, and strainers jalli dipstick for hand-washing.
6. Before carrying out routine cleaning of the tank dipstick should also be removed and put aside carefully.
7. Use long handle brushes for cleaning pipes and hose.
8. Take special attention in the areas like outlet valve, plunger and plunger rod, dipstick socket, corners of the BMC, agitator and paddles (under the paddles) in the BMC during cleaning operation.
9. Rinse with hot (45°C) water. Allow the parts to dry. Assemble all the parts in position.
10. Spray/Circulate cold sanitizer **Prudent 15** solution (0.15%) one hour before taking milk in the BMC tank.
11. Ensure No water is sprayed after cleaning and sanitization.



Food Grade Quality Certificate

Application in food & beverage processing facilities

Here we confirm that, RossClean BMC, if used according to the instructions in the product information sheet is suitable for the open plant cleaning in Beverage, Brewing, Dairy & Food Processing facilities and not to adulterate the food or milk product in any manner during the use.

Essential condition is a thorough rinsing with clear portable water of surfaces intended to get in direct contact with foodstuff.

The remaining residuals that may contaminate or adulterate food contacting cleaned surfaces are expected to be negligible, following a potable water rinse. Therefore, the potential ingestion of residues resulting from the approved use of these types of products does not represent a significant health hazard after our assessment of the products as suitable for use under the intended conditions.

A handwritten signature in blue ink, reading "Manjiri Paranjape".



Thanking you,
Yours faithfully,
For Rossari Biotech Ltd.

Manjiri Paranjape
Senior GM – R&D



Please note that this is not a regulatory assessment and should not be construed as an endorsement of regulatory compliance.

Declaration of Non corrosive and Non toxic

At dilution of 0.3%

Application in Dairy & Food Processing facilities

Here we confirm, that RossClean BMC, if used according to the instructions in the product information sheet is suitable for the cleaning of Dairy & Food Processing facilities and is non-corrosive to metal or human hands and non-toxic if used as per suggested user concentration/dilution in any manner during the use in Bulk Milk Chilling Units.

Toxicological Data

Sr. No	Particular	Acute ORAL (LD ₅₀) [Rat]
1	Poly Hexa Methylene Biogunide	>2000 mg/kg
2	Di Decyl Di Methyl Ammonium Chloride	> 344 mg/kg
3	Benzyl Konium Chloride	> 240 mg/kg



Certificate of Origin

Characteristics Origin Content Official Statement

Product Number	FROBML000K005
Product Name	RossClean BMC
Application	Terminal disinfectant cleaner designed for both CIP and manual cleaning
Always non-animal source?	Yes
Type of Manufacture	Chemical
Sub-Type	Synthetic Organic
Material Country of Origin	INDIA
Country of Manufacture	INDIA

Rossari Biotech Limited's product **RossClean BMC**, containing Alkyldimethyl benzylammonium chloride, Didecyl Dimethyl Ammonium Chloride, Biguanide, Non-anionic Surfactants & other proprietary ingredients, manufactured at its Silvassa facility, are the result of chemical syntheses involving only pure chemical substances.

Rossari products are not of animal origin and do not contain, make use of, or involve, at any point of their manufacturing processes, raw materials of animal origin, including animal proteins.

Rossari's manufacturing facility operates under an ISO 9001 & 14001 Quality System, Green Clean Certified and in compliance with GMP.

Thanking you,

Yours faithfully,
Ms. Manjiri Paranjape
Sr. General Manager
Research & Development



The above information is believed to be correct to the best of our knowledge. This information is to be used for the purpose of determining animal origin only and not to be confused with "country of origin" for import/export purposes. Rossari Biotech Limited shall not be held liable for any damage resulting from handling or from contact with the above product.



Manufacturing Process Flow Chart

Product Number

FROBML000K005

Product Name

RossClean BMC

Application

Terminal disinfectant cleaner designed for both CIP and manual cleaning



Process Steps

- Mix all raw materials in required quantity.
- After adding all raw material mix it for some time.
- Blend it.
- Fill it in a 5 kg jerry cans.
- Put 2 jerry cans in a carton and seal it. Store it under proper ambient conditions as recommended by Rossari Biotech representatives. For more information please refer contact your Rossari Biotech service team.



Total Heavy Metals* Contents

Product	RossClean BMC
Application	Terminal disinfectant cleaner designed for both CIP and manual cleaning
Method	ISO 105 E04:2013
Test Report No	0192413092
Date of Report	03-05-2019

Element's	MDL (mg/kg)	Results	Requirement
Soluble Cobalt (Co)	0.5	< 0.5	-
Soluble Nickel (Ni)	0.5	< 0.5	-

Test result is drawn according to the kind and extent of tests performed.

Without permission of the test centre this test report is not permitted to be duplicated in extracts. This test report does not entitle to carry any safety mark on this or similar products. This test report represents the test parameters as requested by the customer based on submitted samples only.